🚀 GitHub Deployment — הפתרון החינם!

איך להריץ את הלומדה חינם על GitHub

📖 תוכן עניינים

סקירה

שתי אפשרויות חינמיות:

```
┌─────────────────────────────────────────────────────────┐
│         GitHub + Free Hosting = $0/month               │
├─────────────────────────────────────────────────────────┤
│                                                         │
│ 1 אפשרות: Vercel (Frontend) + Railway (Backend)        │
│   Frontend:  Vercel (חינם, מהיר)                       │
│   Backend:   Railway.app ($5 חינם/month בהתחלה)        │
│   DB:        Railway PostgreSQL (חינם)                  │
│   → https://my-app.vercel.app                           │
│   → 5000 :סדר גודל requests/day                         │
│                                                         │
│ 2 אפשרות: GitHub Pages + GitHub Actions                │
│   Frontend:  GitHub Pages (חינם!)                      │
│   Backend:   Railway (חינם)                            │
│   DB:        Railway (חינם)                            │
│   → https://username.github.io/rf-learning             │
│   → ב מובנהGitHub, אין צורך בשרתים                    |
│                                                         │
└─────────────────────────────────────────────────────────┘
```

GitHub Pages + Actions

מה זה GitHub Pages?

GitHub Pages = אחסון אתרים סטטיים **חינם** ישירות מGitHub repository!

```
Your Repository
     ↓
Push to GitHub
     ↓
GitHub Actions (CI/CD)
     ↓
Build Frontend → dist/
     ↓
GitHub Pages
     ↓
https://username.github.io/rf-learning (חינם!)
```

יתרונות GitHub Pages:

✅ **חינם לגמרי** — אין עלויות חודשיות
✅ **טעון ב**version control — **Git** מובנה
✅ **CI/CD אוטומטי** — GitHub Actions
✅ **SSL/HTTPS** — בחינם מטעם GitHub
✅ **דומיין משלך** — אפשר להוסיף custom domain
✅ **Unlimited bandwidth** — ללא הגבלה

חסרונות GitHub Pages:

❌ **רק Frontend סטטי** — צריך Backend אחרת (Railway)
❌ **אין Node.js בצד שלהם** — להריץ backend צריך Railway/Heroku
❌ **אין database** — צריך Railway PostgreSQL

Railway.app (Backend + DB חינם!)

מה זה Railway?

Platform = **Railway** לDeployment עם **חינם $5/month credit** (כמעט חינם!)

```
Git Push
    ↓
Railway (detects Node.js project)
    ↓
Builds automatically
    ↓
Deploys to https://api.railway.app
    ↓
PostgreSQL included (free tier!)
```

יתרונות Railway:

✅ **מעט CLI** — כמעט כמו Heroku
✅ Database — **PostgreSQL included** בחינם
✅ **$5/month credit** — מספיק לרוב בחינם
✅ **Environment variables** — חביב לסודות
✅ **Auto-deploy from Git** — דחוף וזה עובד

מחירים Railway:

```
$5/month Free Credit = enough for:
- 1x Backend (Node.js)
- 1x PostgreSQL Database
- ~1000 API requests/day

לכן = כמעט חינם!
```

Vercel (Frontend חינם)

מה זה Vercel?

Vercel = אחסון Frontend **חינם** (עשה את Next.js)

```
Frontend Build
    ↓
Push to GitHub
    ↓
Vercel (auto-detects Vite)
    ↓
Builds & Deploys
    ↓
https://my-app.vercel.app (חינם!)
```

יתרונות Vercel:

✅ **חינם לגמרי** — אחסון, SSL, bandwidth
✅ **מהיר מאוד** — CDN עולמי
✅ **Auto-deploy from GitHub** — push and done
✅ **Environment variables** — טוב לAPI secrets
✅ **Custom domains** — רק אתה שלך

Setup שלם (Step-by-Step)

שלב 1: סידור GitHub

1.1 צור Repository

```
# Create new repo on GitHub
# Name: rf-learning
# Description: Learning management system
# Public (כדי להשתמש בGitHub Pages)
```

1.2 Push Code to GitHub

```
# In your local project
git init
git add .
git commit -m "Initial commit: rf-learning system"
git branch -M main
git remote add origin https://github.com/YOUR-USERNAME/rf-
learning.git
git push -u origin main
```

After:

```
Your repo is on GitHub!
https://github.com/YOUR-USERNAME/rf-learning
```

שלב 2: Setup Frontend on Vercel

2.1 Connect to Vercel

```
# Option 1: Via Web
# 1. Go to https://vercel.com
# 2. Sign in with GitHub
# 3. Import project → select rf-learning
# 4. Click Deploy

# Option 2: Via CLI
npm install -g vercel
cd frontend
vercel
# Follow prompts
```

2.2 Configure Frontend

Create `frontend/.env.production`:

```
VITE_API_URL=https://api.railway.app
# Later: replace with your Railway backend URL
```

Update `frontend/src/services/api.ts`:

```
const API_URL = process.env.VITE_API_URL ||
'http://localhost:3000/api';

export const apiService = {
  async fetch(endpoint: string, options?: RequestInit) {
    const response = await fetch(`${API_URL}${endpoint}`, options);
    return response.json();
  }
};
```

Deploy Frontend 2.3

```
# Push to GitHub
git add .
git commit -m "Setup Vercel deployment"
git push

# Vercel auto-detects and deploys!
# Check: https://dashboard.vercel.com/projects
```

:Your Frontend is now live at

```
https://rf-learning.vercel.app
```

שלב 3: Setup Backend on Railway

3.1 Create Railway Account

```
1. Go to https://railway.app
2. Sign up with GitHub
3. Authorize
```

3.2 Create New Project

```
1. Dashboard → New Project
2. Deploy from GitHub repo
3. Select: rf-learning
4. Choose: backend directory
```

3.3 Add PostgreSQL

```
1. Add Service → Add from Marketplace
2. Select PostgreSQL
3. Click Add
```

3.4 Configure Environment Variables

In Railway Dashboard:

```
Backend service:
NODE_ENV = production
SERVER_PORT = 3000
DB_HOST = (auto-filled from PostgreSQL)
DB_PORT = (auto-filled)
DB_USER = postgres
DB_PASSWORD = (auto-generated)
DB_NAME = rf_learning
JWT_SECRET = your-super-secret-key-change-this
UPLOAD_DIR = /tmp/uploads
```

3.5 Railway Auto-Deploy

```
Push to GitHub → Railway detects → Auto builds → Auto deploys!
```

Your Backend is now live at:

```
https://api.railway.app/api/courses
(Railway gives you exact URL)
```

שלב 4: Connect Frontend to Backend

4.1 Get Backend URL from Railway

```
Railway Dashboard → Backend Service → Domain
Copy: https://rf-learning-api.railway.app
```

4.2 Update Frontend .env

`frontend/.env.production`:

```
VITE_API_URL=https://rf-learning-api.railway.app/api
```

4.3 Redeploy Frontend

```
git add frontend/.env.production
git commit -m "Update API URL for Railway backend"
git push
# Vercel auto-redeploys!
```

שלב 5: Database Setup on Railway

5.1 Connect to PostgreSQL

```
# Get connection string from Railway
# Railway Dashboard → PostgreSQL → Database URL

# Copy the URL, looks like:
# postgresql://user:pass@host:port/dbname
```

5.2 Create Tables

Option A: Via Railway Console

```
1. Railway Dashboard → PostgreSQL
2. Click "Connect" → Query
3. Paste SQL from backend/src/db.sql
4. Run
```

Option B: Via CLI

```
psql "postgresql://user:pass@host:port/dbname" < backend/src/db.sql
```

5.3 Verify Database

```
psql "postgresql://user:pass@host:port/dbname"

# Check tables
\dt

# Should show: users, courses, chapters, learning_content, etc.
```

שלב 6: CI/CD Pipeline (GitHub Actions)

6.1 Auto-Run Tests on Push

`.github/workflows/test.yml` :Create

```yaml
name: Run Tests

on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - uses: actions/setup-node@v3
        with:
          node-version: '18'

      - name: Install dependencies
        run: npm install

      - name: Build Frontend
        run: cd frontend && npm run build

      - name: Build Backend
        run: cd backend && npm run build
```

Auto-Deploy on Merge to Main 6.2

```
# Railway auto-deploys when you push to main
# No extra setup needed!
```

Comparison — עלויות

Free Tier Comparison

Service	Free	Cost
GitHub	Yes ✅	$0
Vercel Frontend	Yes ✅	$0
Railway Backend	Yes ($5 credit/month) ✅	$0~
Railway PostgreSQL	Yes ✅	$0
SSL/HTTPS	Yes ✅	$0
Custom Domain	Yes ✅	$0
Monthly Total		**!$0**

הצעות ניתוחים עתידיות (Upscale)

כשאתה גדל:

```
Tier 1 (0-1000 users): FREE
  - Vercel Free
  - Railway $5/month
  - GitHub Free
  Total: $0-5/month

Tier 2 (1000-10,000 users): $20-50/month
  - Vercel Pro ($20)
  - Railway upgrade ($25)
  - GitHub Pro ($4)
  Total: $49/month

Tier 3 (10,000+ users): $100+/month
  - Move to AWS/DigitalOcean
  - Load balancing
  - CDN optimization
```

להפעלה — Quick Checklist

```
☐ GitHub Repository created & code pushed
☐ Vercel connected & Frontend deployed
☐ Railway connected & Backend deployed
☐ PostgreSQL database created on Railway
☐ Tables created (db.sql ran)
☐ Environment variables set
☐ Backend URL updated in Frontend
☐ Frontend redeployed
☐ Test: Frontend loads at https://rf-learning.vercel.app
☐ Test: API responds at https://api.railway.app/api/courses
☐ Test: Login works
☐ Test: View courses works
```

יתרונות GitHub Deployment

בהשוואה לDigitalOcean ($5/month):

```
DigitalOcean           GitHub + Vercel + Railway
─────────────────────────────────────────────────
$5/month               $0-5/month
צריך SSH               Git push only
צריך DevOps            Fully automated
צריך backup            Git IS backup
צריך מוניטורינג        Built-in
שליטה מלאה             Less control
```

יתרונות GitHub Deployment:

✅ **חינם לגמרי** (עם Railway credit)
✅ **Git-based** — כל פעם שאתה push‏דש, זה מדפיס
✅ **CI/CD אוטומטי** — GitHub Actions
✅ **Version control built-in** — rollback קל
✅ **Collaboration** — שתף עם team
✅ **Comments on commits** — דוקומנטציה
✅ **Pull requests** — code review

Deployment Flow Diagram

```
┌─────────────────────────────────────────────────────┐
│          You Work Locally                            │
│  npm run dev, make changes, test                     │
└─────────────────────────────────────────────────────┘
                      ↓
                 git push origin main
                      ↓
        ┌───────────────────────────────┐
        │    GitHub Receives Push        │
        │  (monitoring both frontend/    │
        │   backend directories)         │
        └───────────────────────────────┘
              ↓                      ↓
    ┌─────────┐              ┌──────────┐
    │ Vercel  │              │ Railway  │
    ├─────────┤              ├──────────┤
    │ Detected│              │ Detected │
    │frontend/│              │backend/  │
    │build    │              │build     │
    └────┬────┘              └────┬─────┘
         │                        │
    ┌────▼────┐              ┌────▼─────┐
    │ npm run │              │ npm run   │
    │ build   │              │ dev       │
    │         │              │ + PM2     │
    └────┬────┘              └────┬─────┘
         │                        │
    ┌────▼────────────────────────▼────┐
    │  Tests run (GitHub Actions)        │
    └────┬──────────────────────────────┘
         │ ✅ if pass
```

```
┌────▼───────────────────────────────────┐
│  Deploy to Production                   │
│  Frontend: Vercel                       │
│  Backend: Railway                       │
└──────────────────────────────────┘
     │
┌────▼──────────────────────────────┐
│  Live! 🎉                          │
│  https://app.vercel.app           │
│  https://api.railway.app          │
└──────────────────────────────┘
```

פקודות בסיסיות — Day-to-Day

```
# כל יום, אתה רק עושה:
git add .
git commit -m "Fix login bug"
git push

# The rest is automatic! 🎉
# Vercel + Railway handle deployment
# GitHub Actions handle testing
```

Troubleshooting

"Frontend can't connect to Backend"

```
1. Check Railway backend URL is correct
2. Update frontend/.env.production
3. Push to GitHub
4. Vercel will rebuild automatically
```

"Database not found on Railway"

```
1. Go to Railway Dashboard
2. PostgreSQL service → Connect
3. Paste connection string in Railway Backend env vars
4. Run db.sql via Railway console
```

"Build fails on Vercel"

```
1. Check: npm run build works locally
2. Check: frontend/package.json has all dependencies
3. Check: No hardcoded paths or localhost references
4. Push again (Vercel will retry)
```

סיכום — התוכנית המוצלחת

המדריך הקצר ביותר:

```
Step 1: git push your code to GitHub
Step 2: Connect Vercel (5 minutes)
Step 3: Connect Railway (5 minutes)
Step 4: Add PostgreSQL on Railway (1 minute)
Step 5: Create tables (1 minute)
Step 6: Update API URL in Frontend (1 minute)
Step 7: Push again
Step 8: Done! ✅

Total time: 20 minutes
Cost: $0/month

Your app is live and auto-updating!
```

Free Hosting + Git Version Control = 🚀

```
GitHub Pages (Frontend)     ← Free
Railway (Backend + DB)      ← Free
Auto-Deploy on Push         ← Free
SSL/HTTPS                   ← Free
Version Control             ← Free
CI/CD Pipeline              ← Free
```

כל מה שצריך לך = FREE! ✅

עדכון: 2026-06-28

גרסה: 1.0

עלות: $0/month

סטטוס: Production Ready

ספר המערכת - לומדת ענף 71

עדכון אחרון: 2026-06-28

גרסה: Extended 1.0